radxa

Radxa ROCK 4A

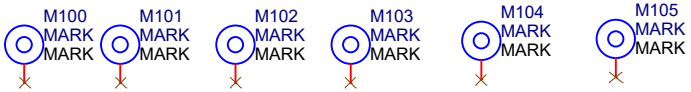
schematic

V1.52

CONTENT INDEXING

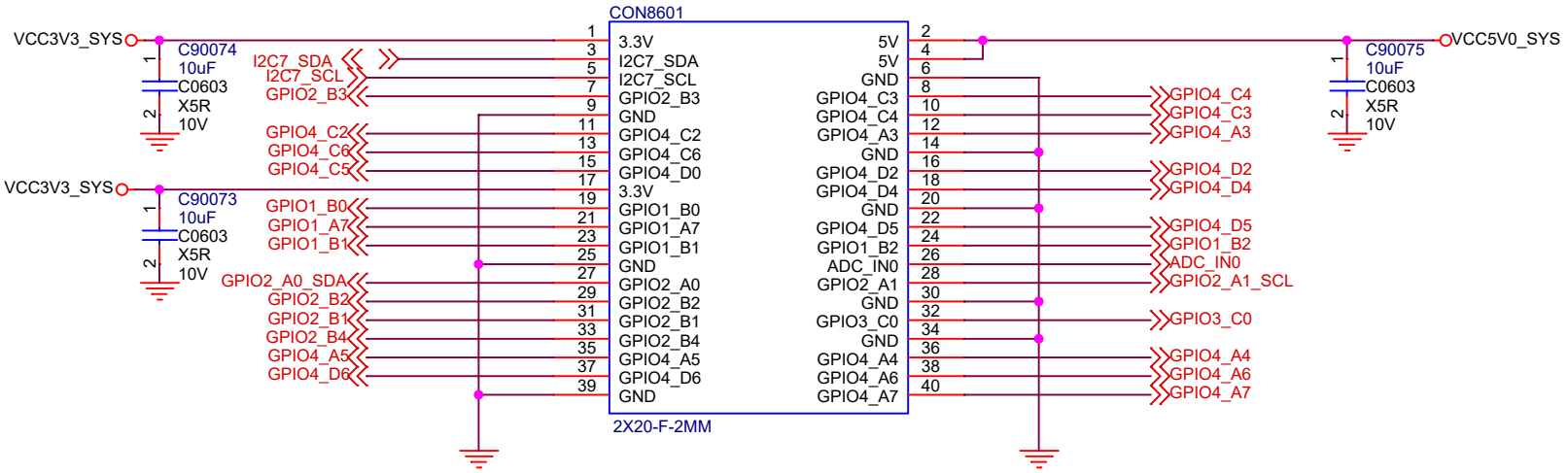
01.Cover
02.Index
03.Connector
04.ES8316
05.PCIe NGFF/M.2
06.RK3399 Power
07.RK3399 PMU Controler
08.RK3399 DDR Controler
09.RK3399 Flash&SDMMC Controler
10.RK3399 USB/USIC Controler
11.RK3399 SARADC/Key
12.RK3399 DVP Interface
13.RK3399 Display Interface
14.RK3399 GPIO
15.RK3399 PCIE
16.Power-DC IN
17.Power-PMIC RK808-D
18.RTL8211F
19.USB3.0 OTG Port
20.RAM-LPDDR4(200P)
21.Memory-EMMC
22.WIFI/BT AP6255
23.TF Card
24.HDMI Output

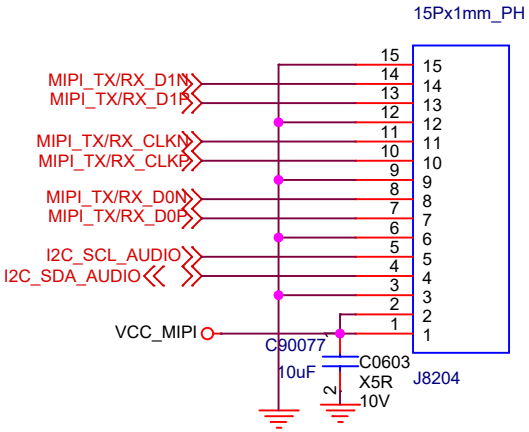


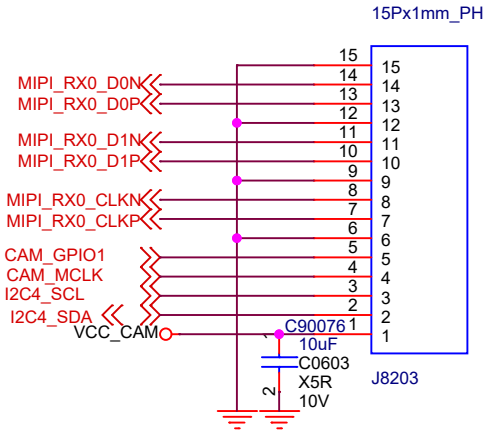


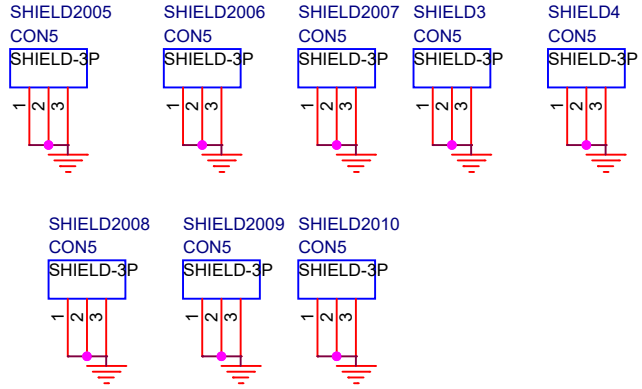
radxa

Size	Title:	REV
A4	ROCK 4A	
	Page Name: Index	1.52
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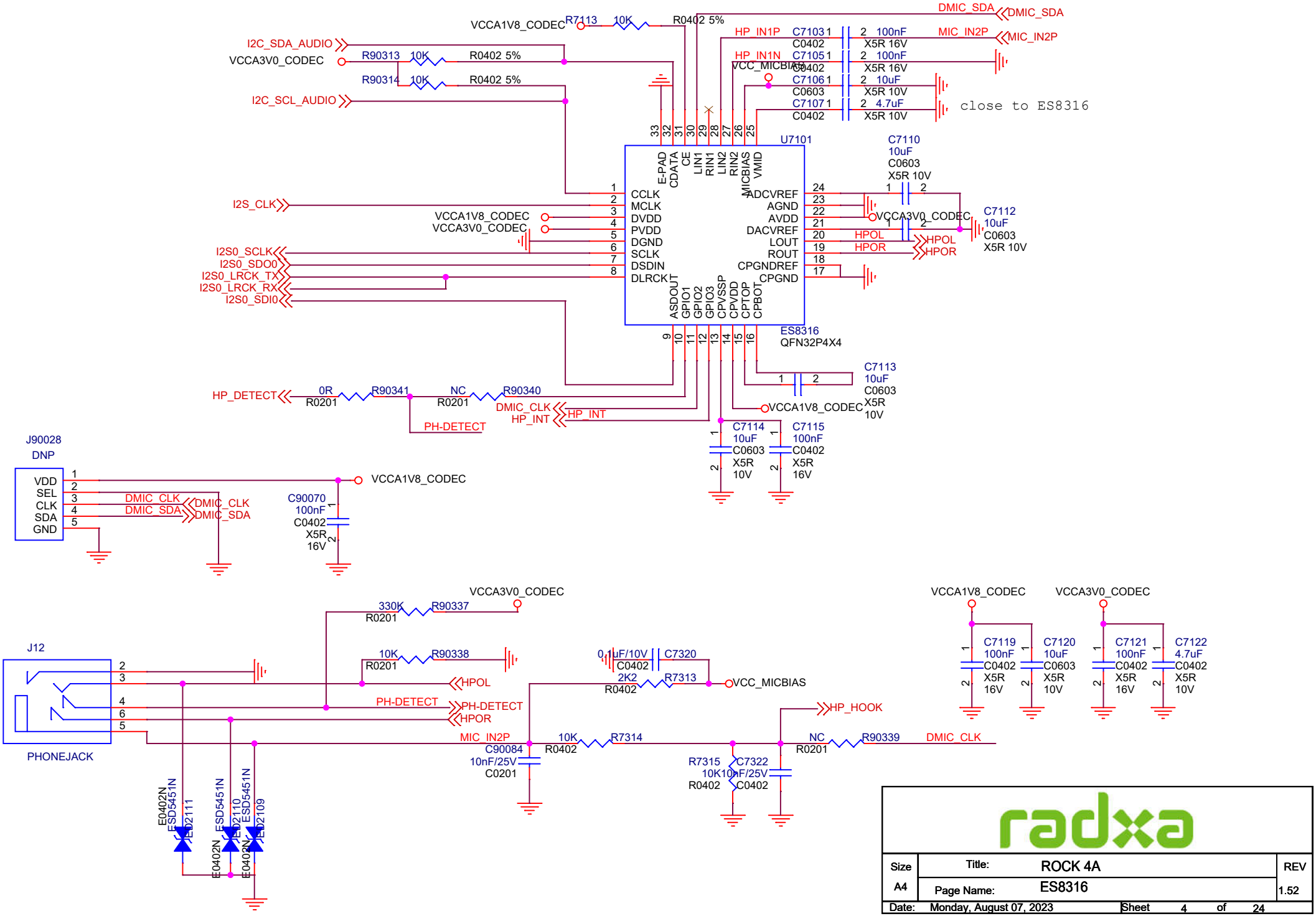




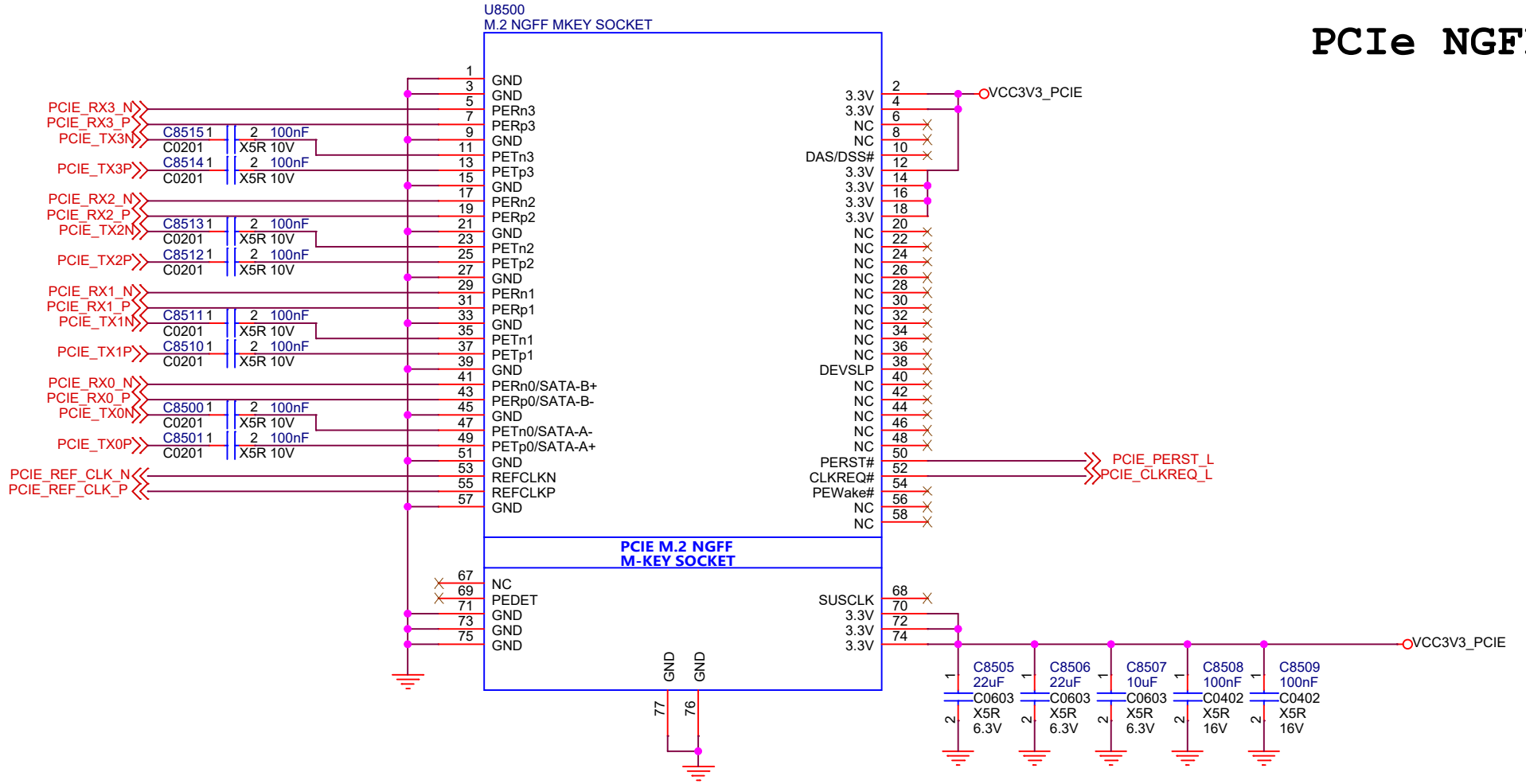


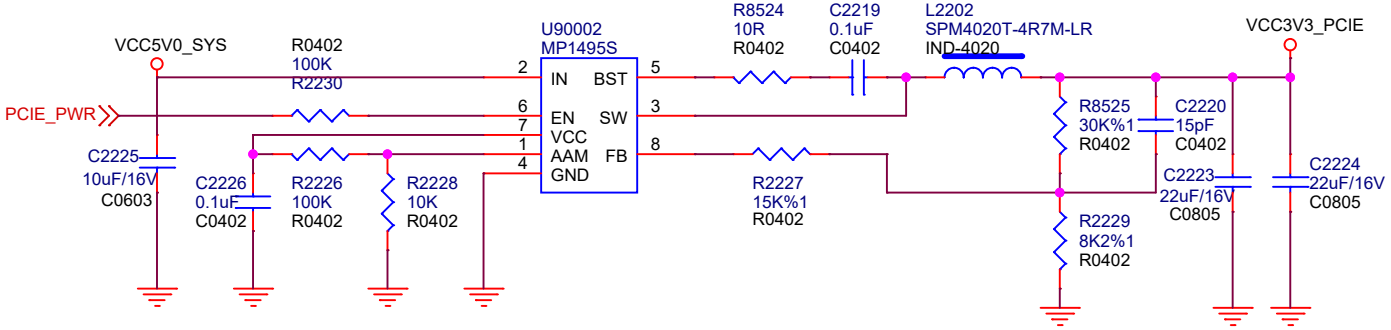


Size A4	Title: ROCK 4A	REV 1.52
	Page Name: CONNECTOR	
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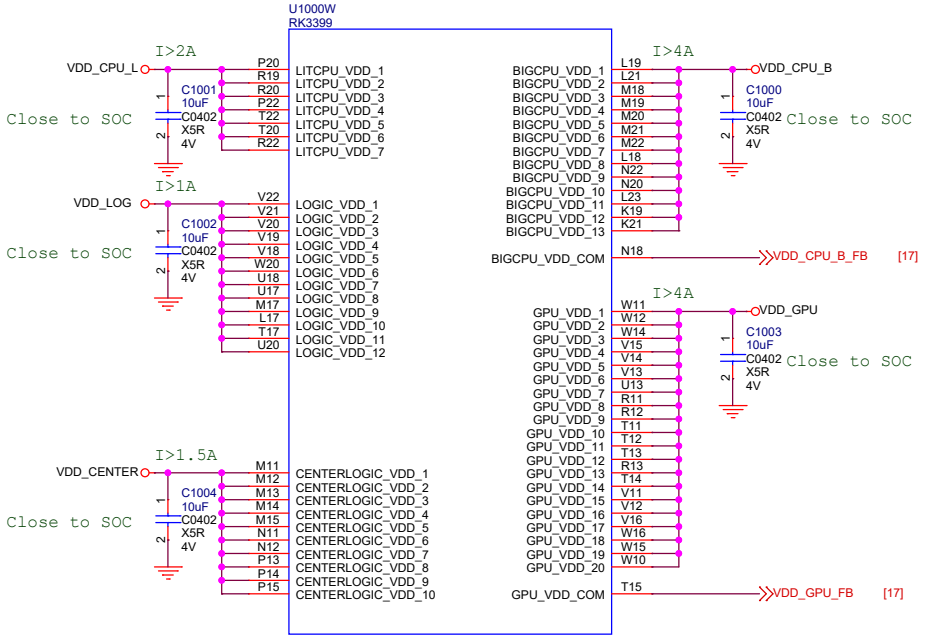

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R7113 10K R0402 5%
DMIC_SDA
I2C_SDA_AUDIO
VCCA3V0_CODEC
R90313 10K R0402 5%
R90314 10K R0402 5%
I2C_SCL_AUDIO
HP_IN1P C7103 1 2 100nF C0402 X5R 16V
MIC_IN2P
HP_IN1N C7105 1 2 100nF C0402 X5R 16V
VCC_MICBIAS
C7106 1 2 10uF C0603 X5R 10V
C7107 1 2 4.7uF C0402 X5R 10V
close to ES8316
U7101
E-PAD CDATA CE LIN1 RIN1 LIN2 RIN2 MICBIAS VMID
CCLK MCLK DVDD PVDD DGND SCLK DSDIN DLRCK
ADCVREF AGND AVDD DACVREF LOUT ROUT CPGNDREF CPGND
ASDOUT GPIO1 GPIO2 GPIO3 CPVSSP CPVDD CPTOP CPBOT
ES8316
QFN32P4X4
C7110 10uF C0603 X5R 10V
C7112 10uF C0603 X5R 10V
I2S_CLK
I2S0_SCLK
I2S0_SDO0
I2S0_LRCK_TX
I2S0_LRCK_RX
I2S0_SDI0
HPOL
HPOR
HP_DETECT
0R R0201 R90341
NC R0201 R90340
PH-DETECT
DMIC_CLK
HP_INT
C7113 10uF C0603 X5R 10V
C7114 10uF C0603 X5R 10V
C7115 100nF C0402 X5R 16V
J90028
DNP
VDD SEL CLK SDA GND
C90070 100nF C0402 X5R 16V
J12
PHONEJACK
330K R0201 R90337
10K R0201 R90338
0.1uF/10V C7320 C0402
2K2 R0402 R7313
HP_HOOK
MIC_IN2P
C90084 10nF/25V C0201
10K R0402 R7314
NC R0201 R90339
R7315 10K R0402
C7322 10nF/25V C0402
E0402N ESD5451N LED2111
E0402N ESD5451N LED2110
E0402N ESD5451N LED2109
C7119 100nF C0402 X5R 16V
C7120 10uF C0603 X5R 10V
C7121 100nF C0402 X5R 16V
C7122 4.7uF C0402 X5R 10V
radxa
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PCIe NGFF/M.2

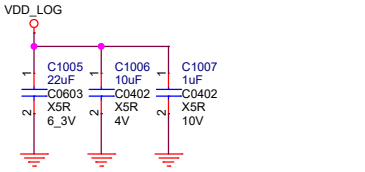


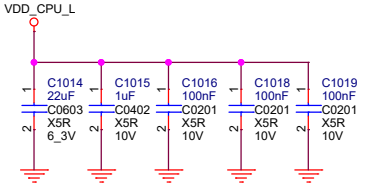


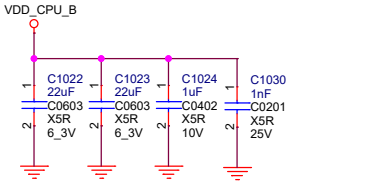
Size A4	Title: ROCK 4A		REV 1.52
	Page Name: PCIe NGFF/M.2		
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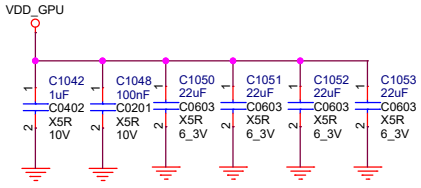


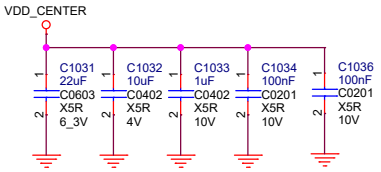
Note:Power filter CAP please place back of SOC or close to SOC

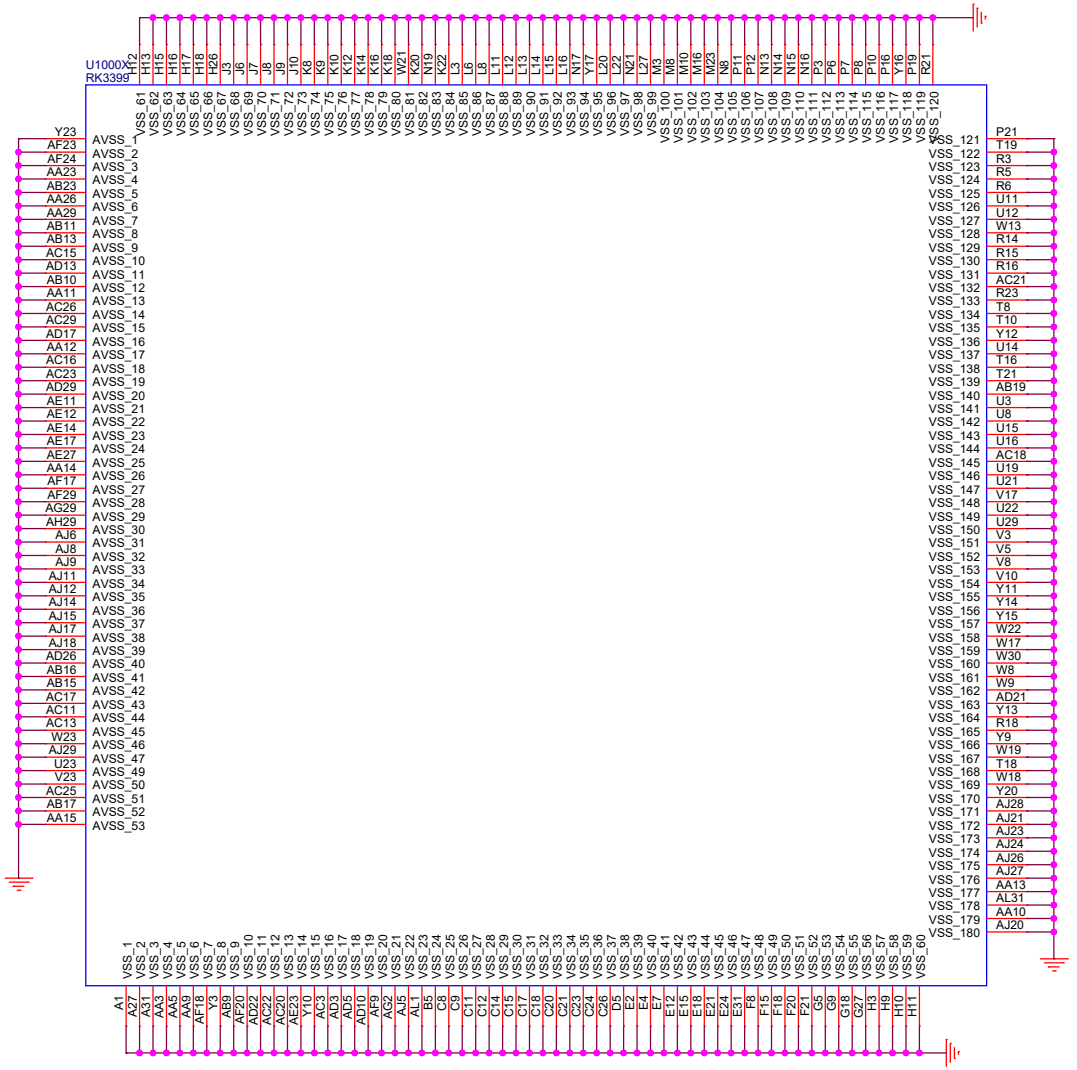




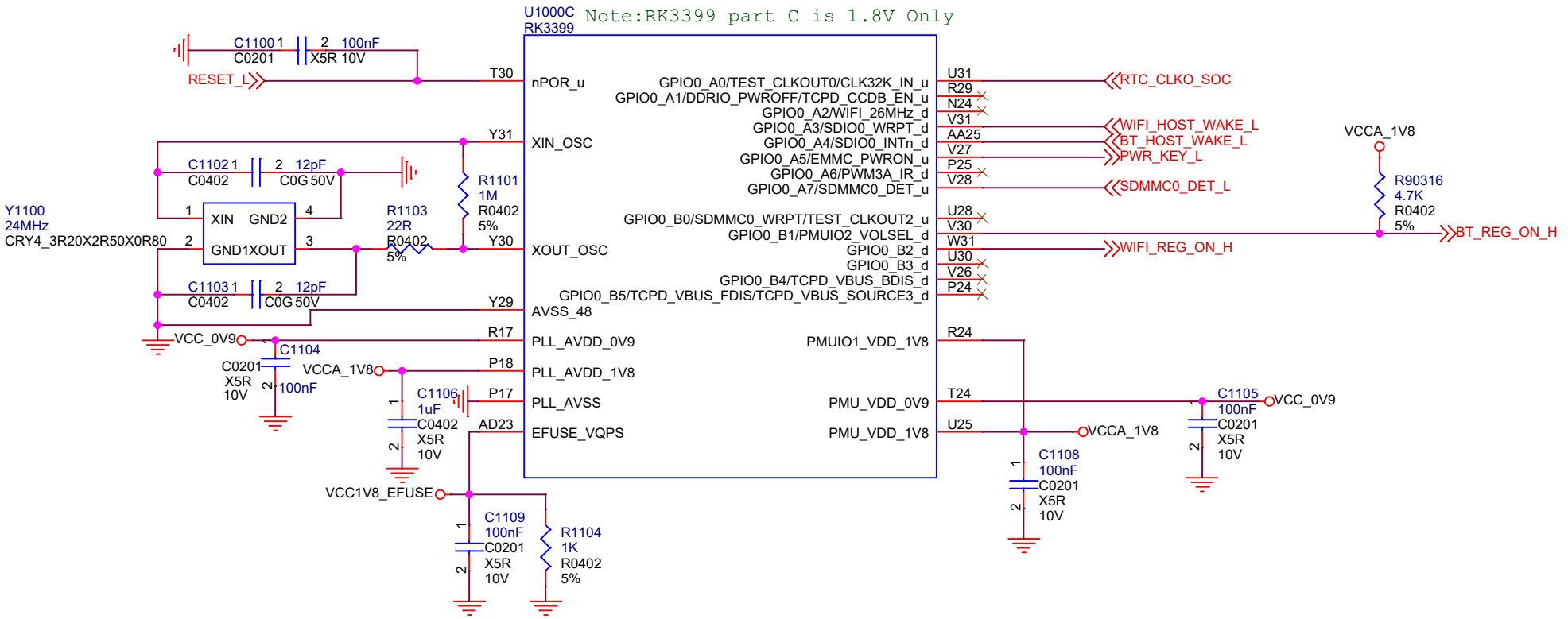




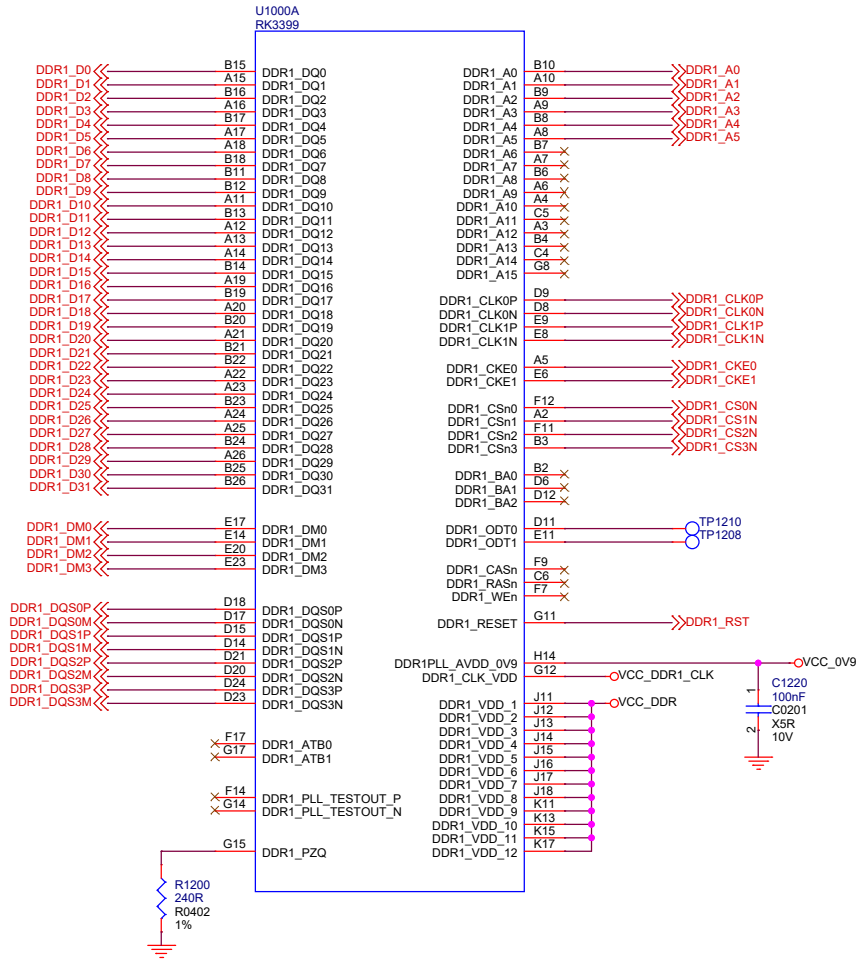


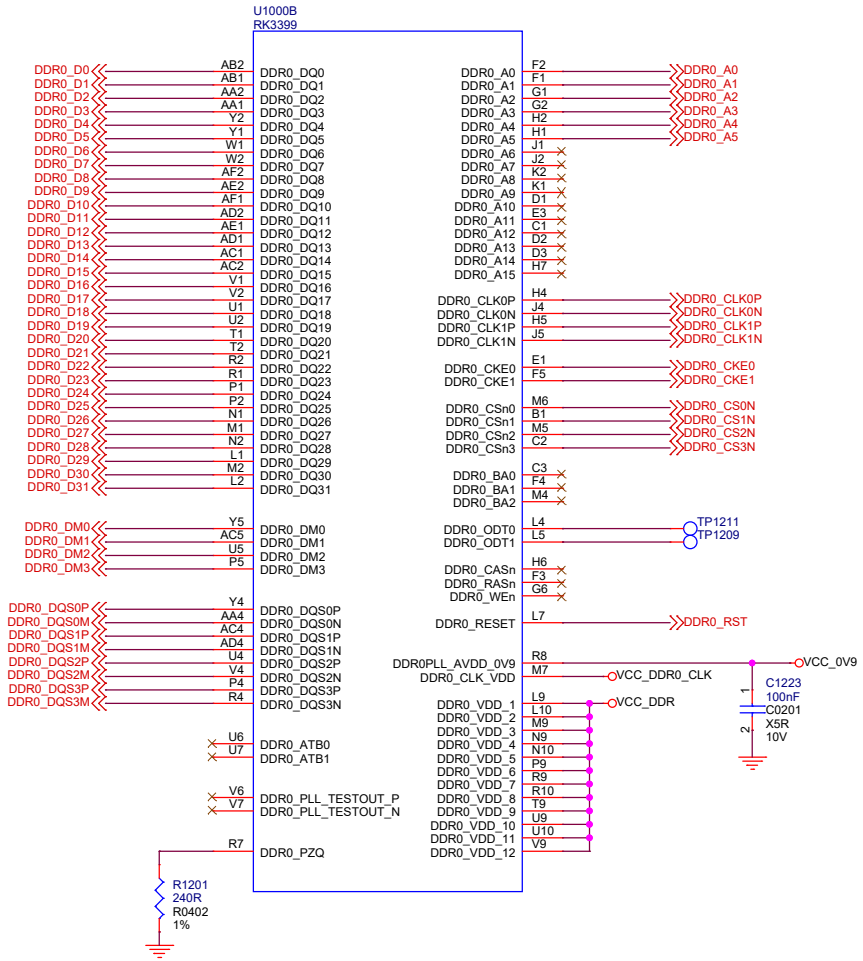


Size	Title: ROCK 4A		REV
A3	Page Name: RK3399 Power		1.52
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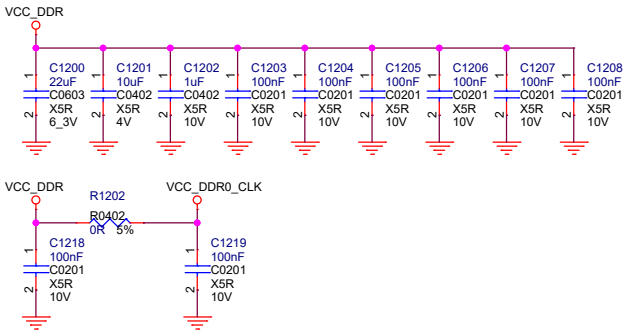


Size A4	Title: ROCK 4A	REV 1.52
	Page Name: RK3399 PMU Controler	
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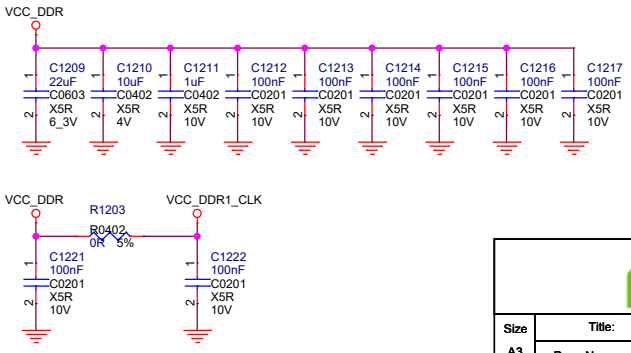




DDR FILTER Note:R1202 cannot be deleted

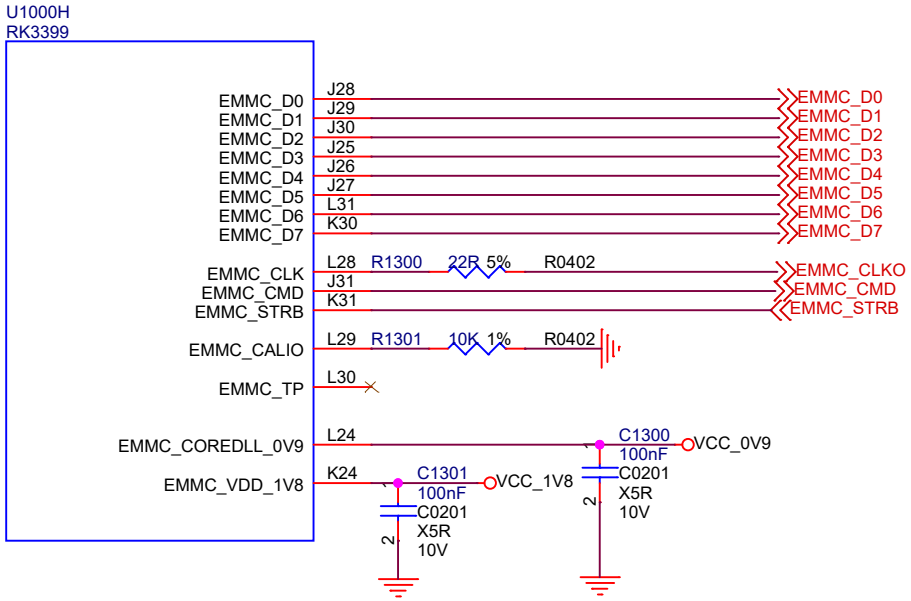


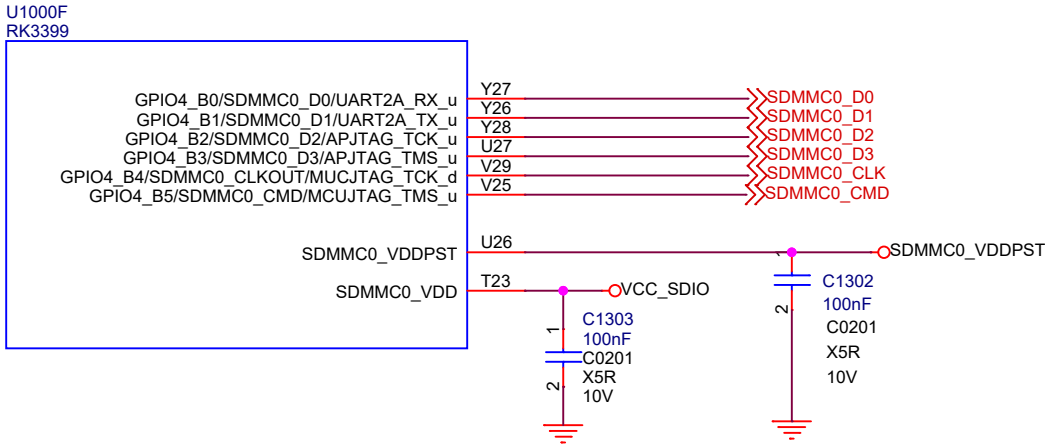
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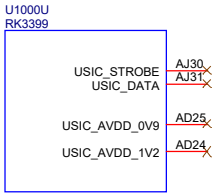
radxa

Size	Title:	REV
A3	ROCK 4A	1.52
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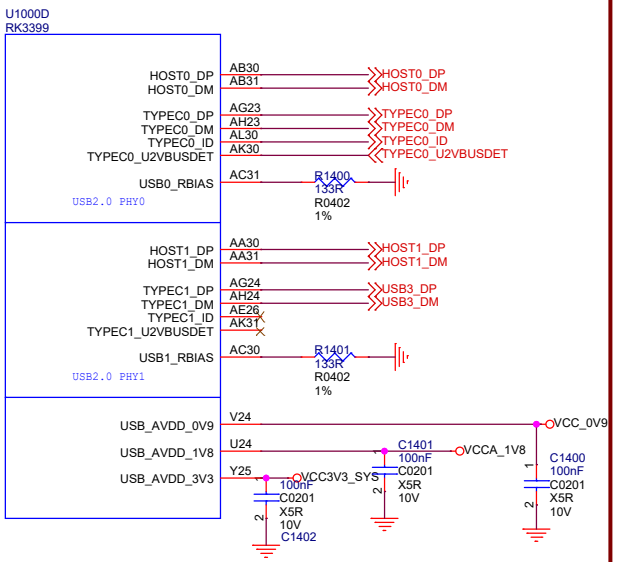




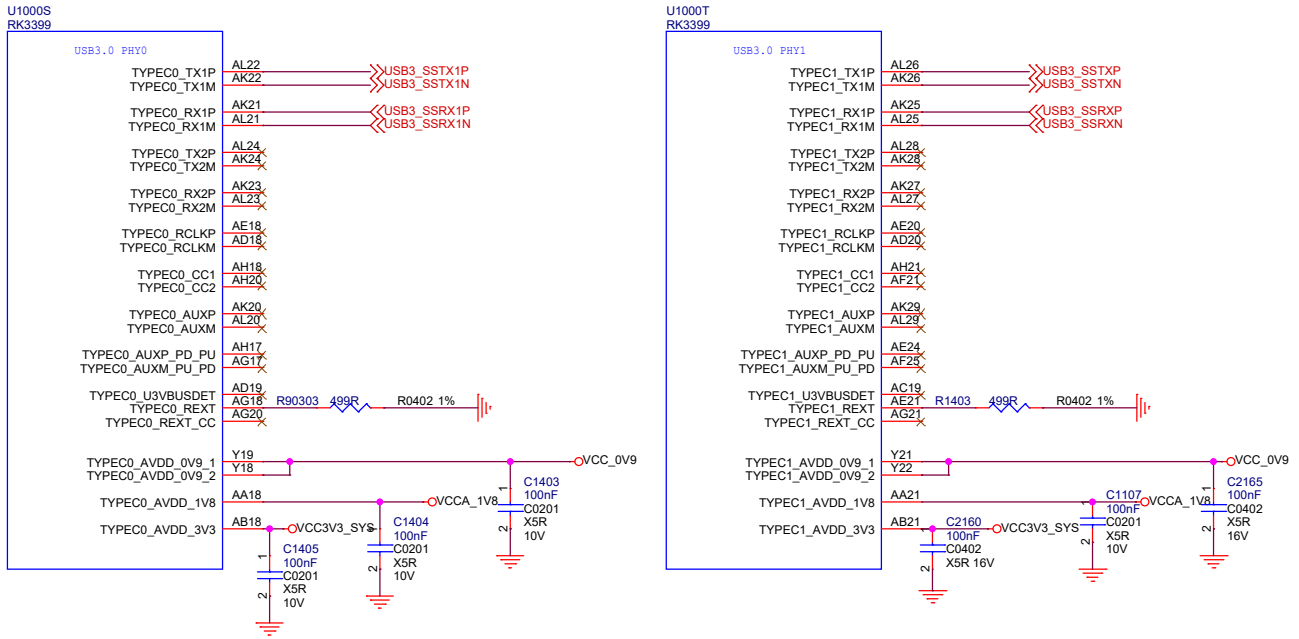
Size A4	Title: ROCK 4A	REV 1.52
	Page Name: RK3399 Flash&SDMMC Controler	
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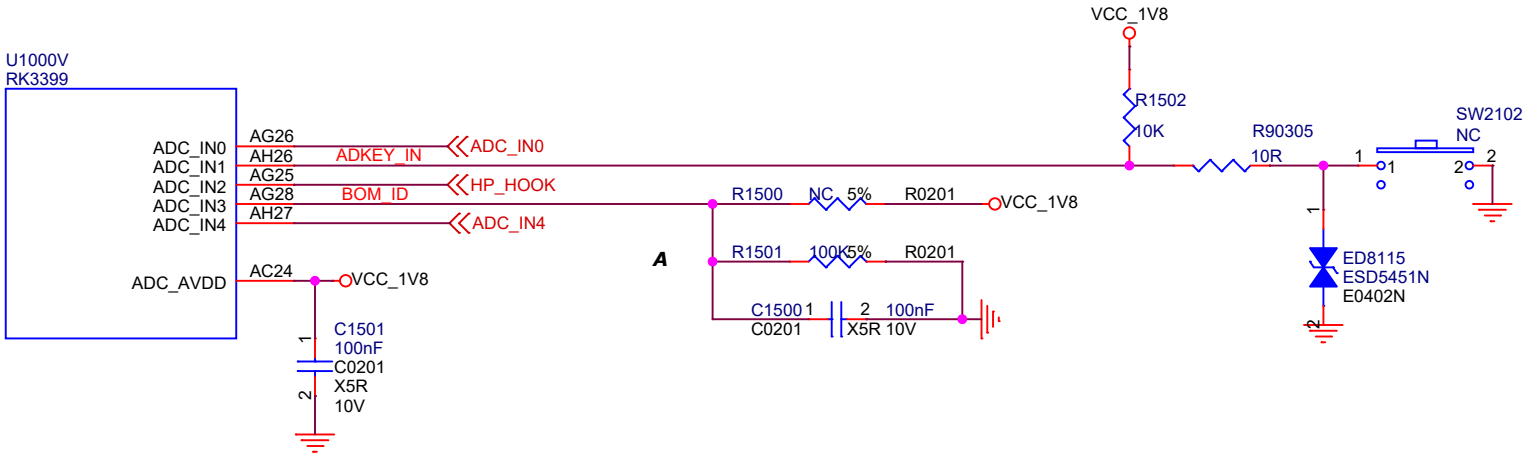
USB2.0



USB3.0



Size: A3	Title: ROCK 4A Page Name: RK3399 USB/USIC Controler	REV: 1.52
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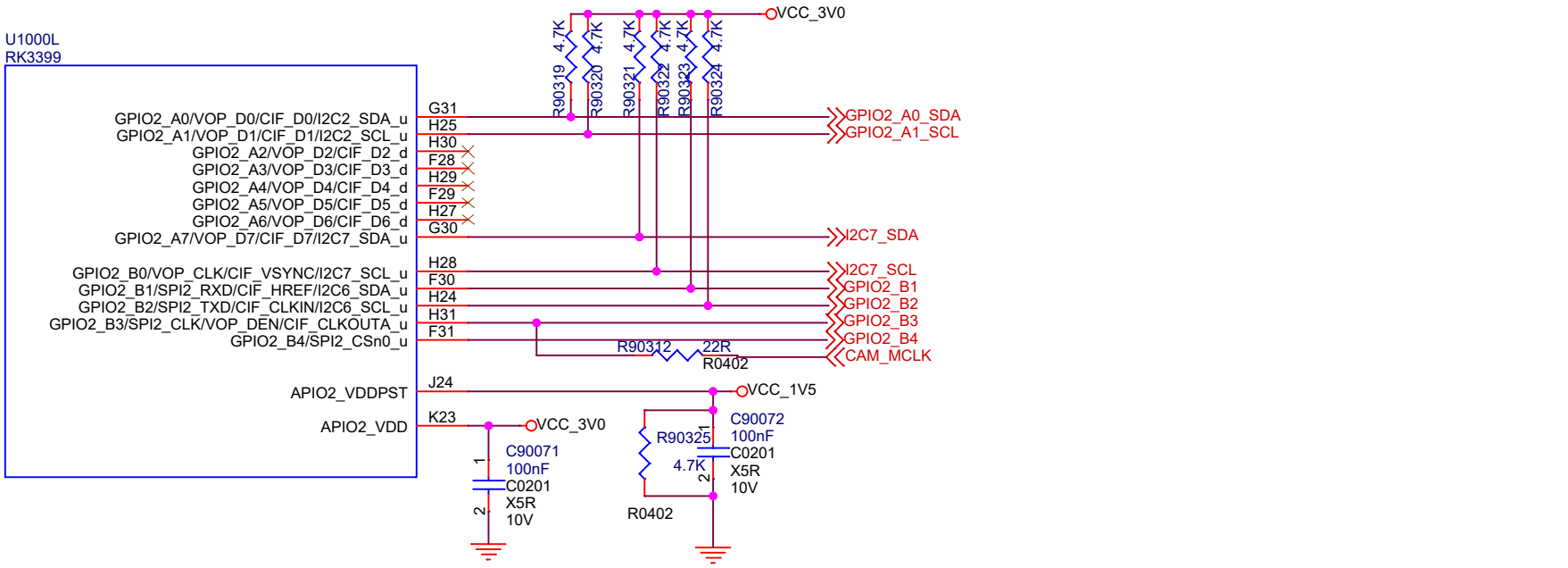
KEY BAORD

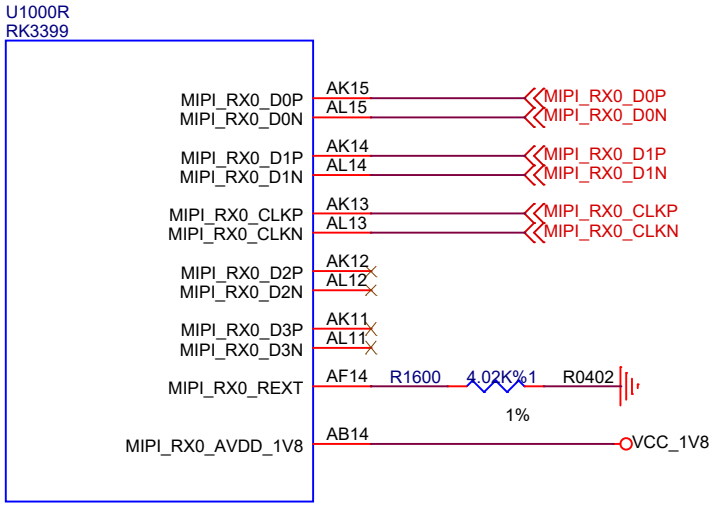
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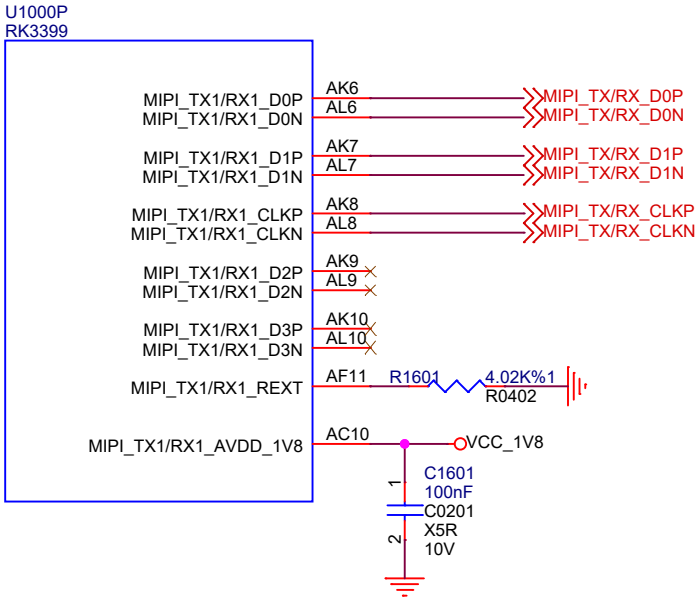
Item	Rup	Rdown		VOL	VERSION
LEVEL1	DNP	100K		0V	A
LEVEL2	100K	20K		0.3V	B
LEVEL3	100K	51K		0.6V	C
LEVEL4	100K	100K		0.9V	D
LEVEL5	100K	200K		1.2V	E
LEVEL6	100K	499K		1.5V	F
LEVEL7	100K	DNP		1.8V	G

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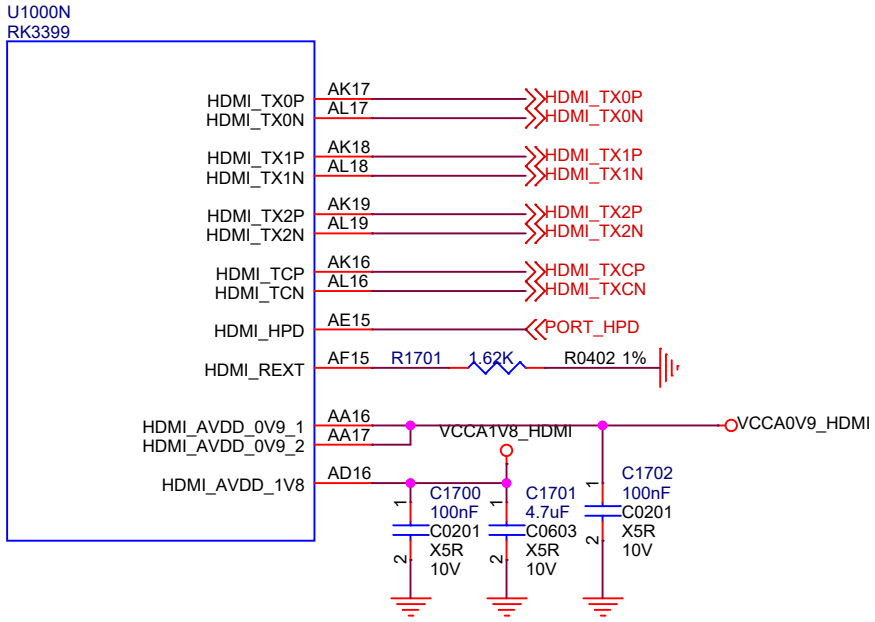
Size	Title:	ROCK 4A	REV
A4	Page Name:	RK3399 SARADC/Key	1.52
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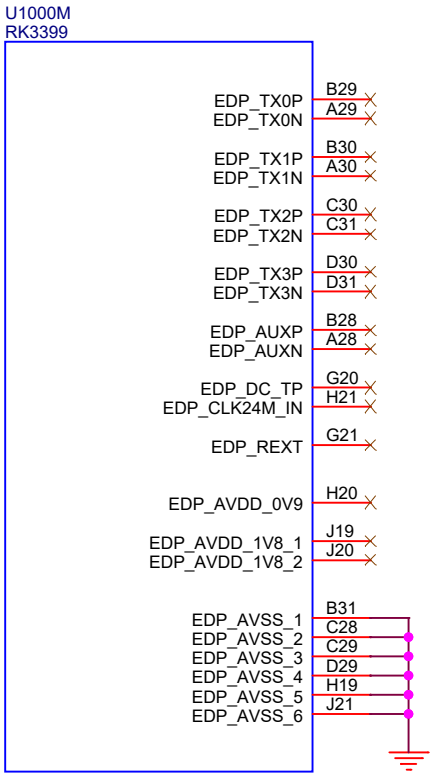


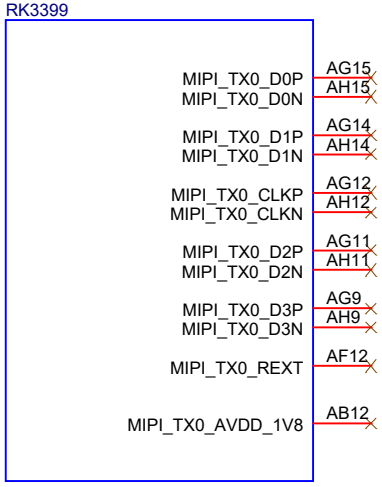




Size A4	Title: ROCK 4A	REV 1.52
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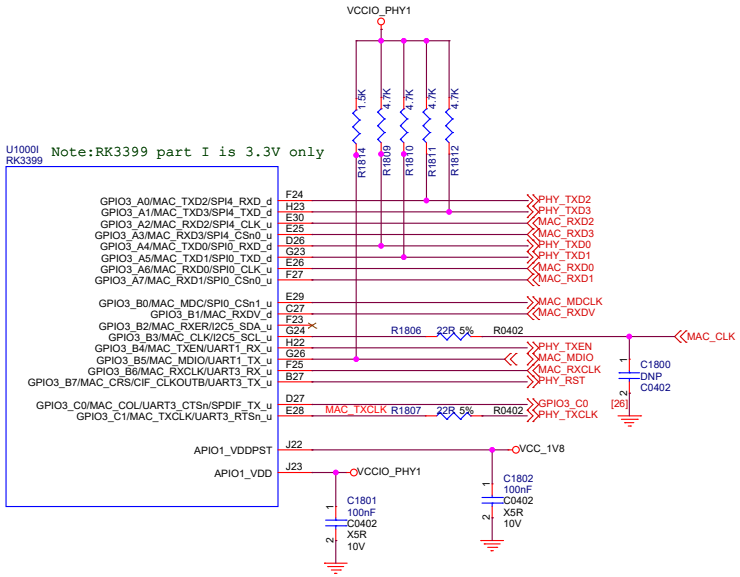


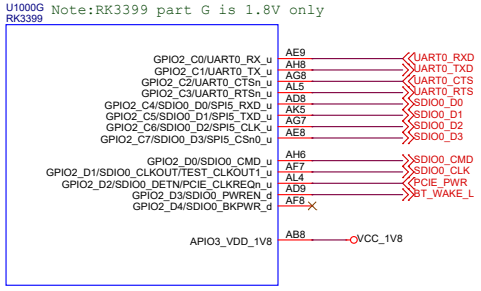


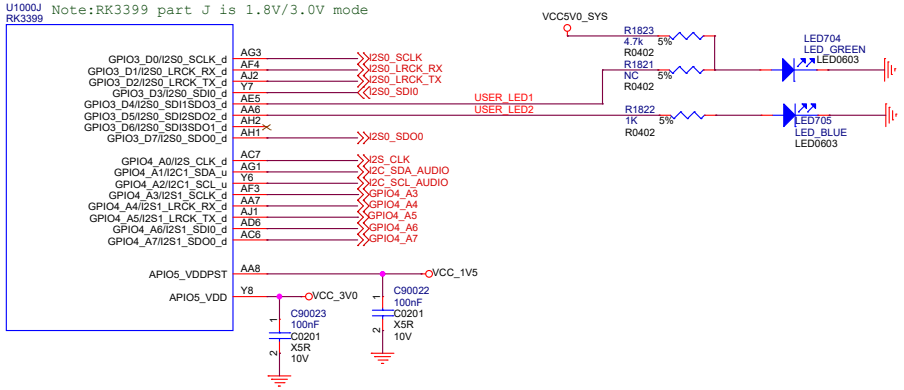


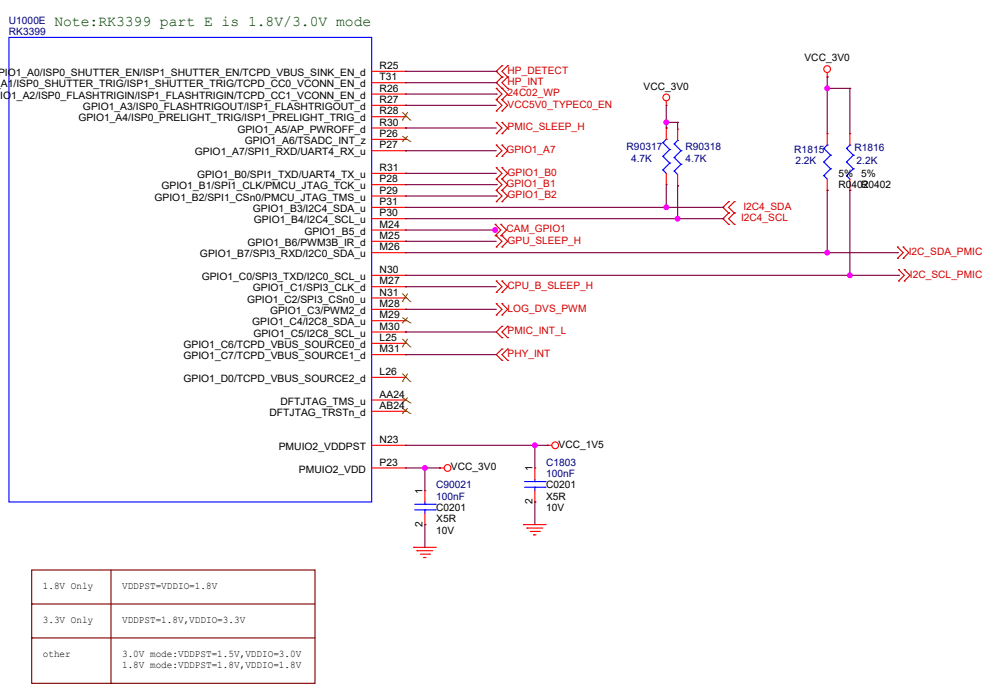
radxa

Size	Title:	REV
A4	ROCK 4A	
	Page Name: RK3399 Display Interface	1.52
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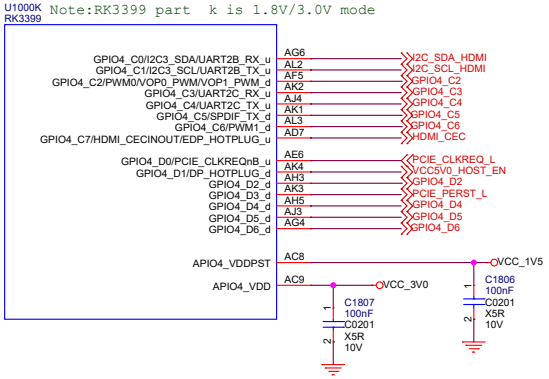






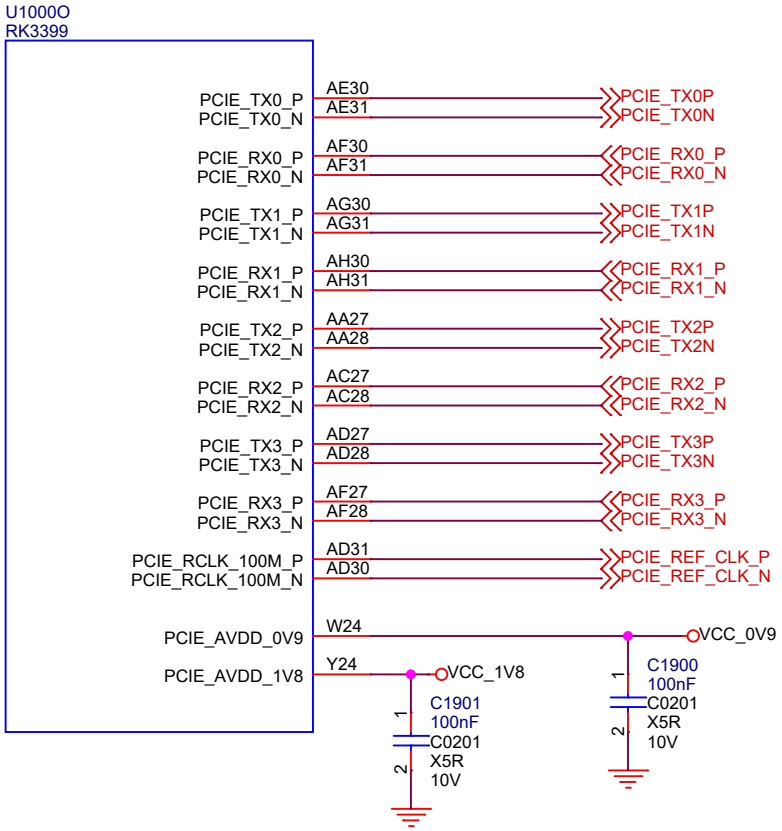


1.8V Only	VDDPST=VDDIO=1.8V
3.3V Only	VDDPST=1.8V,VDDIO=3.3V
other	3.0V mode:VDDPST=1.5V,VDDIO=3.0V 1.8V mode:VDDPST=1.8V,VDDIO=1.8V



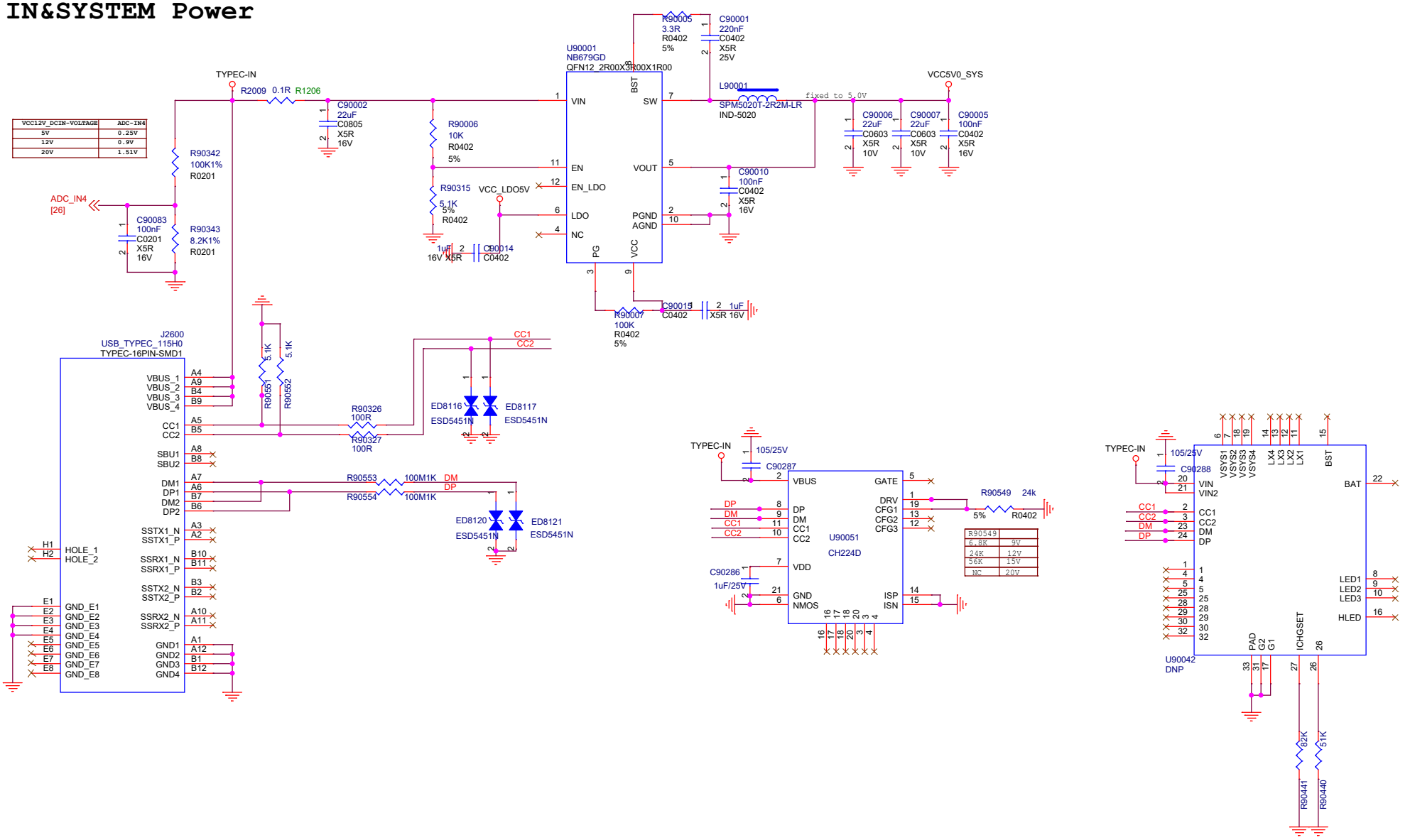
radxa

Size	Title: ROCK 4A	REV
Custom	Page Name: RK3399 GPIO	1.52
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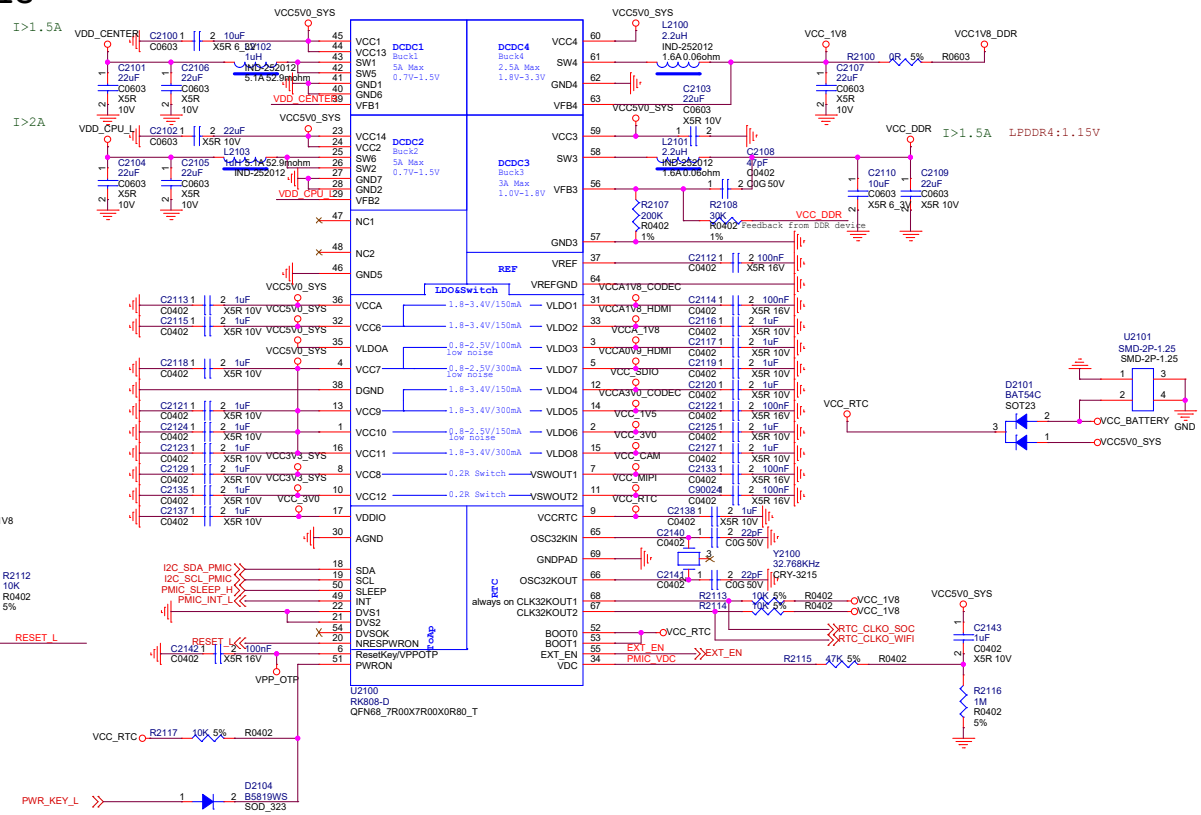
Size A4	Title: ROCK 4A	REV 1.52
	Page Name: RK3399 PCIE	
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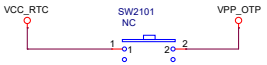
DC IN&SYSTEM Power



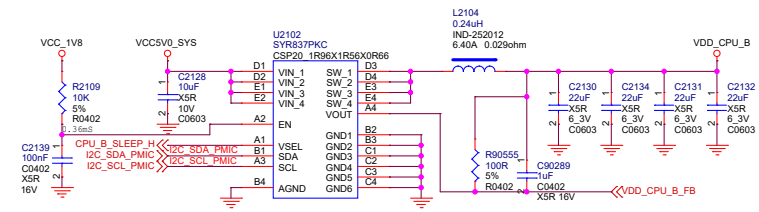
Size	Title: ROCK 4A		REV
A3	Page Name: Power-DC IN		1.52
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PMIC

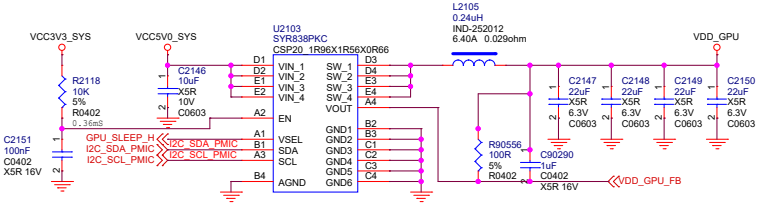




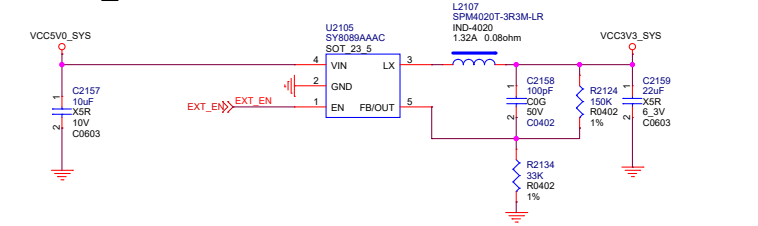
VDD_CPU_B power



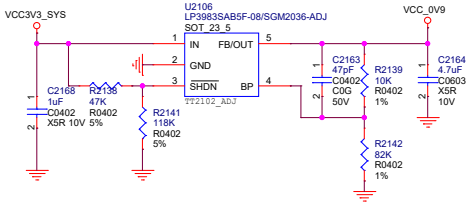
VDD_GPU power



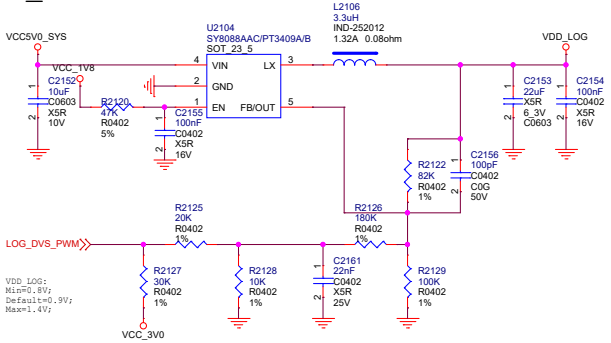
VCC3V3_SYS power

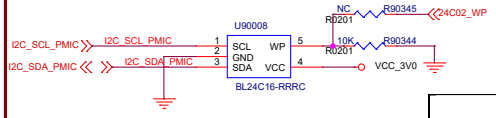


VCC_0V9 power

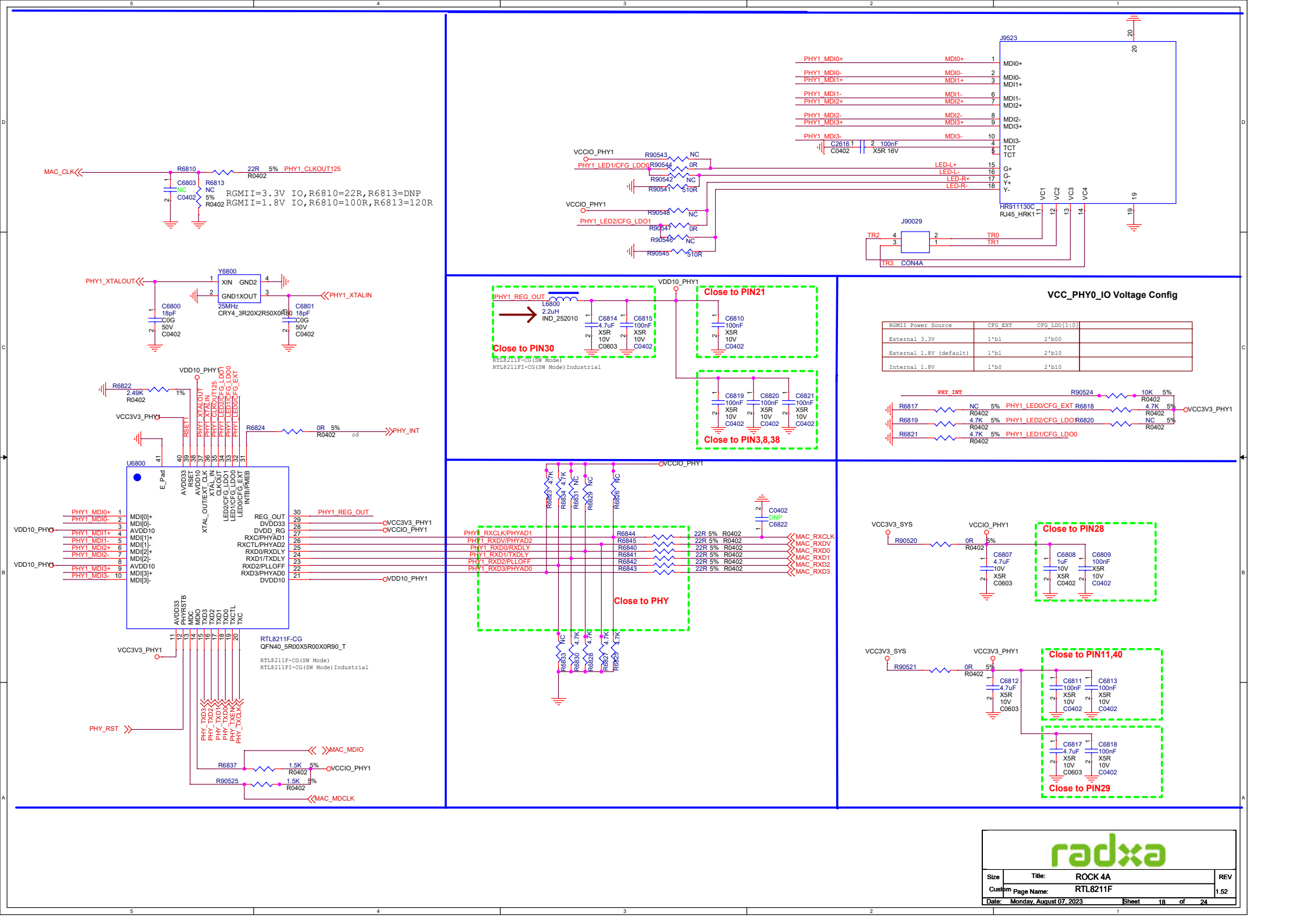


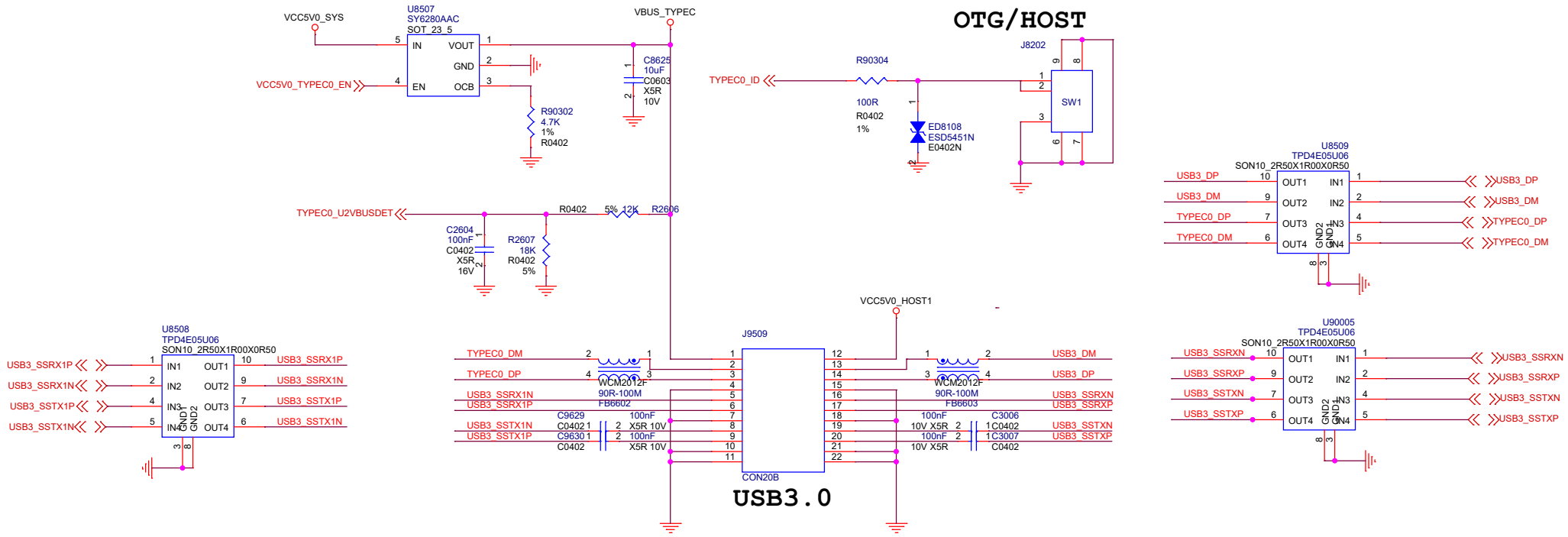
VDD_LOG power



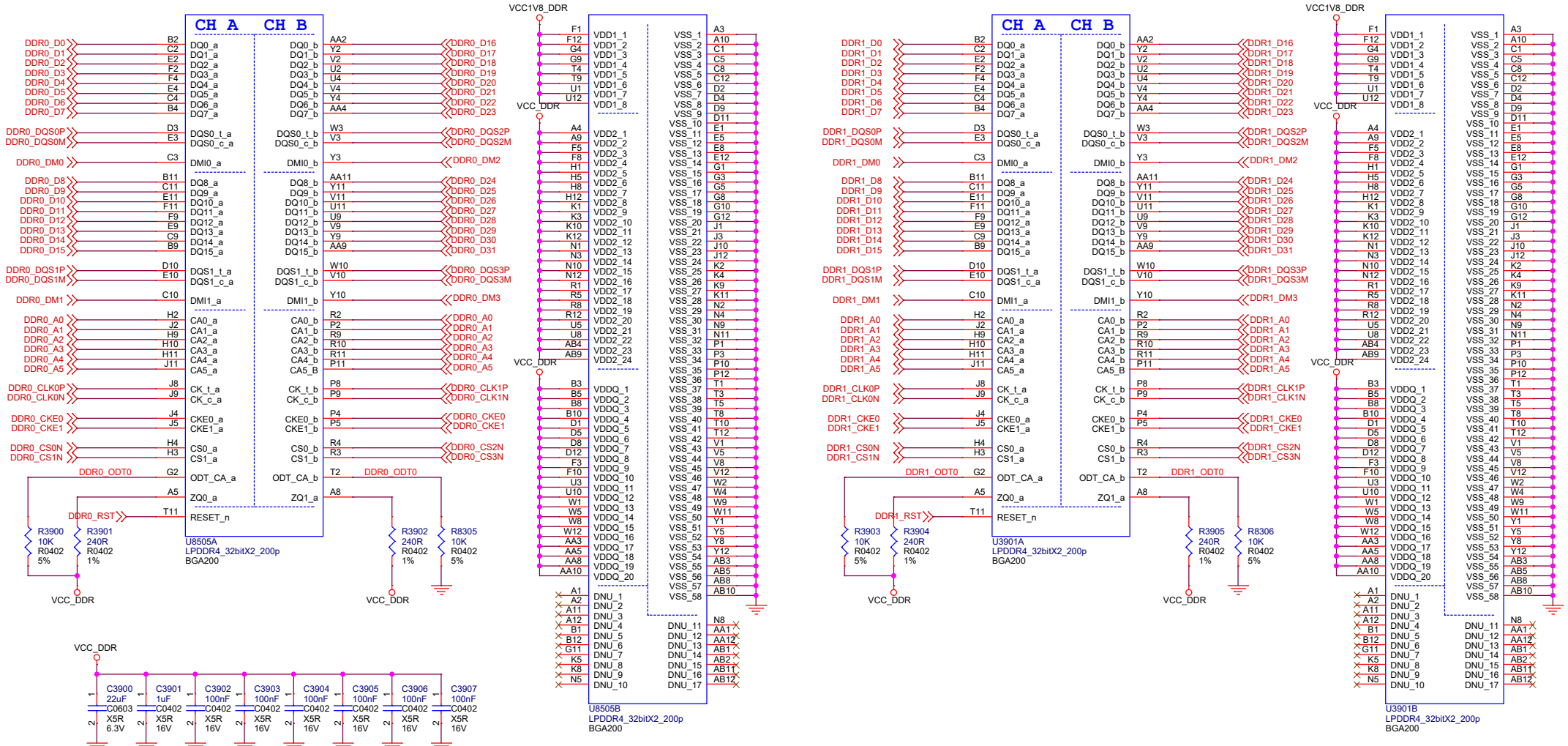


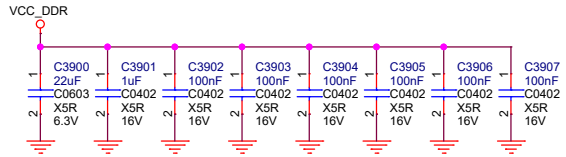
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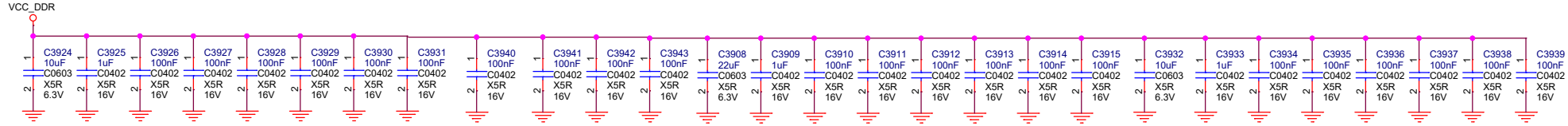

MAC_CLK
R6810
22R 5% PHY1_CLKOUT125
R0402
C6803
NC
C0402
R6813
NC
5%
R0402
RGMII=3.3V IO,R6810=22R,R6813=DNP
RGMII=1.8V IO,R6810=100R,R6813=120R
PHY1_XTALOUT
Y6800
XIN GND2
GND1XOUT
25MHz
CRY4_3R20X2R50X0R80
PHY1_XTALIN
C6800
18pF
C0G
50V
C0402
C6801
18pF
C0G
50V
C0402
VDD10_PHY1
R6822
2.49K
1%
R0402
VCC3V3_PHY
R6824
0R 5%
R0402
PHY_INT
U6800
E_Pad
RSET
AVDD33
AVDD10
XTAL_OUT/EXT_CLK
XTAL_IN
CLKOUT
LED2/CFG_LDO1
LED1/CFG_LDO0
LED0/CFG_EXT
INTB/PMEB
PHY1_MDI0+
PHY1_MDI0-
PHY1_MDI1+
PHY1_MDI1-
PHY1_MDI2+
PHY1_MDI2-
PHY1_MDI3+
PHY1_MDI3-
MDI[0]+
MDI[0]-
AVDD10
MDI[1]+
MDI[1]-
MDI[2]+
MDI[2]-
AVDD10
MDI[3]+
MDI[3]-
REG_OUT
DVDD33
DVDD_RG
RXC/PHYAD1
RXCTL/PHYAD2
RXD0/RXDLY
RXD1/TXDLY
RXD2/PLLOFF
RXD3/PHYAD0
DVDD10
PHY1_REG_OUT
VCC3V3_PHY1
VCCIO_PHY1
VDD10_PHY1
AVDD33
PHYRSTB
MDC
MDIO
TXD3
TXD2
TXD1
TXD0
TXCTL
TXC
RTL8211F-CG
QFN40_5R00X5R00X0R90_T
RTL8211F-CG(SW Mode)
RTL8211FI-CG(SW Mode)Industrial
VCC3V3_PHY1
PHY_RST
PHY_TXD3
PHY_TXD2
PHY_TXD1
PHY_TXD0
PHY_TXEN
PHY_TXCLK
MAC_MDIO
R6837
1.5K 5%
R0402
VCCIO_PHY1
R90525
1.5K 5%
R0402
MAC_MDCLK
J9523
PHY1_MDI0+
PHY1_MDI0-
PHY1_MDI1+
PHY1_MDI1-
PHY1_MDI2+
PHY1_MDI2-
PHY1_MDI3+
PHY1_MDI3-
C2616
C0402
100nF
X5R 16V
MDI0+
MDI0-
MDI1+
MDI1-
MDI2+
MDI2-
MDI3+
MDI3-
TCT
TCT
G+
G-
Y+
Y-
LED-L+
LED-L-
LED-R+
LED-R-
HR911130C
RJ45_HRK1
VC1
VC2
VC3
VC4
VCCIO_PHY1
R90543
NC
PHY1_LED1/CFG_LDO0
R90544
0R
R90542
NC
R90541
510R
VCCIO_PHY1
R90548
NC
PHY1_LED2/CFG_LDO1
R90547
0R
R90546
NC
R90545
510R
J90029
TR2
TR3
CON4A
TR0
TR1
PHY1_REG_OUT
L6800
2.2uH
IND_252010
C6814
4.7uF
X5R
10V
C0603
C6815
100nF
X5R
10V
C0402
VDD10_PHY1
Close to PIN21
C6810
100nF
X5R
10V
C0402
Close to PIN30
RTL8211F-CG(SW Mode)
RTL8211FI-CG(SW Mode)Industrial
C6819
100nF
X5R
10V
C0402
C6820
100nF
X5R
10V
C0402
C6821
100nF
X5R
10V
C0402
Close to PIN3,8,38
VCC_PHY0_IO Voltage Config
RGMII Power Source
CFG_EXT
CFG_LDO[1:0]
External 3.3V
1'b1
2'b00
External 1.8V (default)
1'b1
2'b10
Internal 1.8V
1'b0
2'b10
PHY_INT
R6817
NC 5%
R0402
R6819
4.7K 5%
R0402
R6821
4.7K 5%
R0402
R90524
10K 5%
R0402
PHY1_LED0/CFG_EXT
R6818
4.7K 5%
R0402
PHY1_LED2/CFG_LDO1
R6820
NC 5%
R0402
PHY1_LED1/CFG_LDO0
VCC3V3_PHY1
VCCIO_PHY1
R6823
4.7K
R6834
4.7K
R6831
NC
R6829
NC
R6826
NC
C0402
DNP
C6822
PHY1_RXCLK/PHYAD1
PHY1_RXDV/PHYAD2
PHY1_RXD0/RXDLY
PHY1_RXD1/TXDLY
PHY1_RXD2/PLLOFF
PHY1_RXD3/PHYAD0
R6844
R6845
R6840
R6841
R6842
R6843
22R 5% R0402
MAC_RXCLK
MAC_RXDV
MAC_RXD0
MAC_RXD1
MAC_RXD2
MAC_RXD3
Close to PHY
R6833
NC
R6830
4.7K
R6828
4.7K
R6827
4.7K
R6825
4.7K
VCC3V3_SYS
R90520
0R 5%
R0402
VCCIO_PHY1
C6807
4.7uF
10V
X5R
C0603
Close to PIN28
C6808
1uF
10V
X5R
C0402
C6809
100nF
X5R
10V
C0402
VCC3V3_SYS
R90521
0R 5%
R0402
VCC3V3_PHY1
C6812
4.7uF
X5R
10V
C0603
Close to PIN11,40
C6811
100nF
X5R
10V
C0402
C6813
100nF
X5R
10V
C0402
C6817
4.7uF
X5R
10V
C0603
C6818
100nF
X5R
10V
C0402
Close to PIN29
radxa
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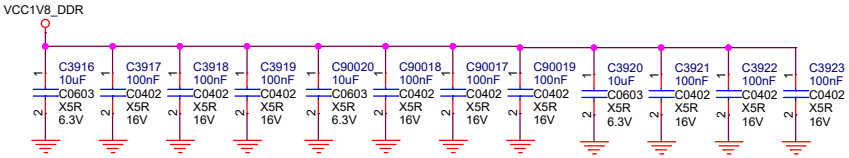


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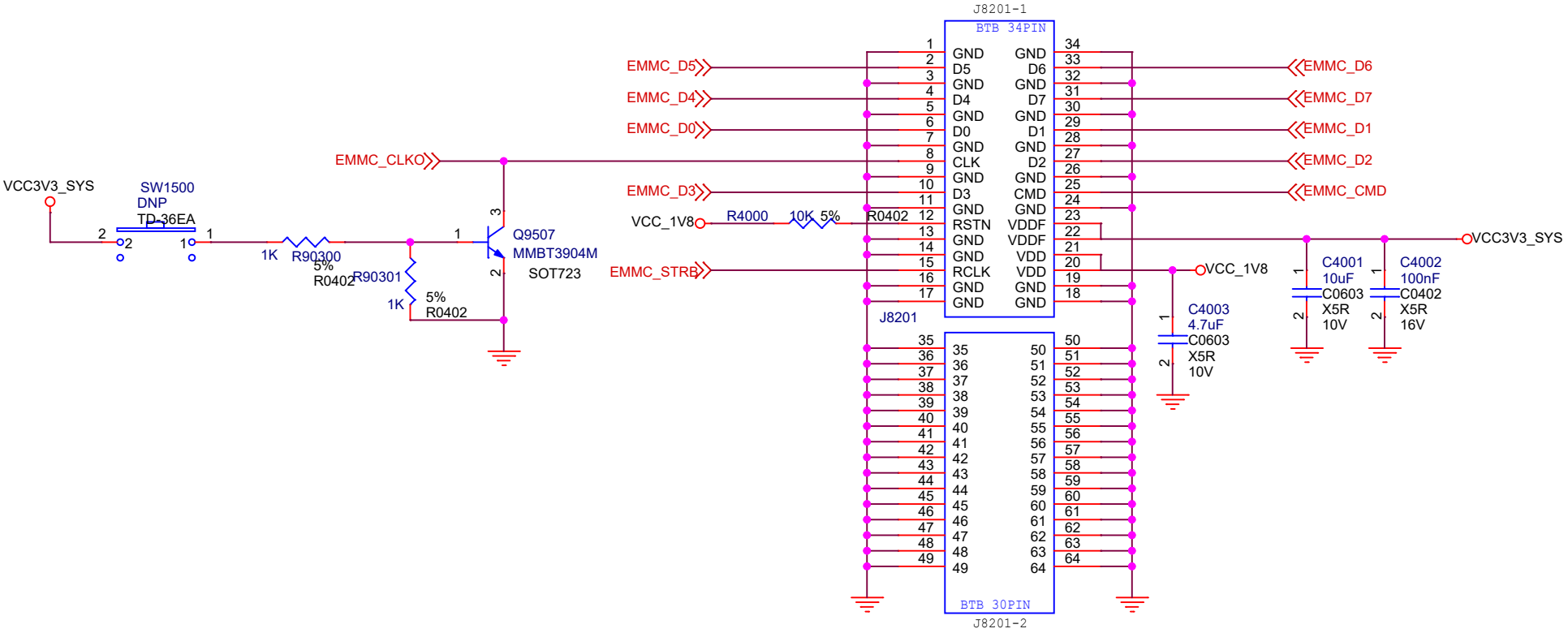


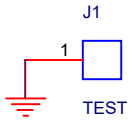


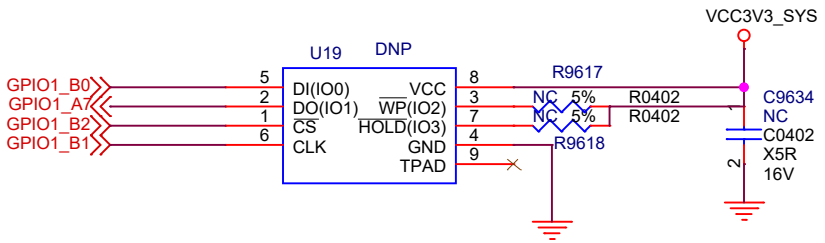
radxa

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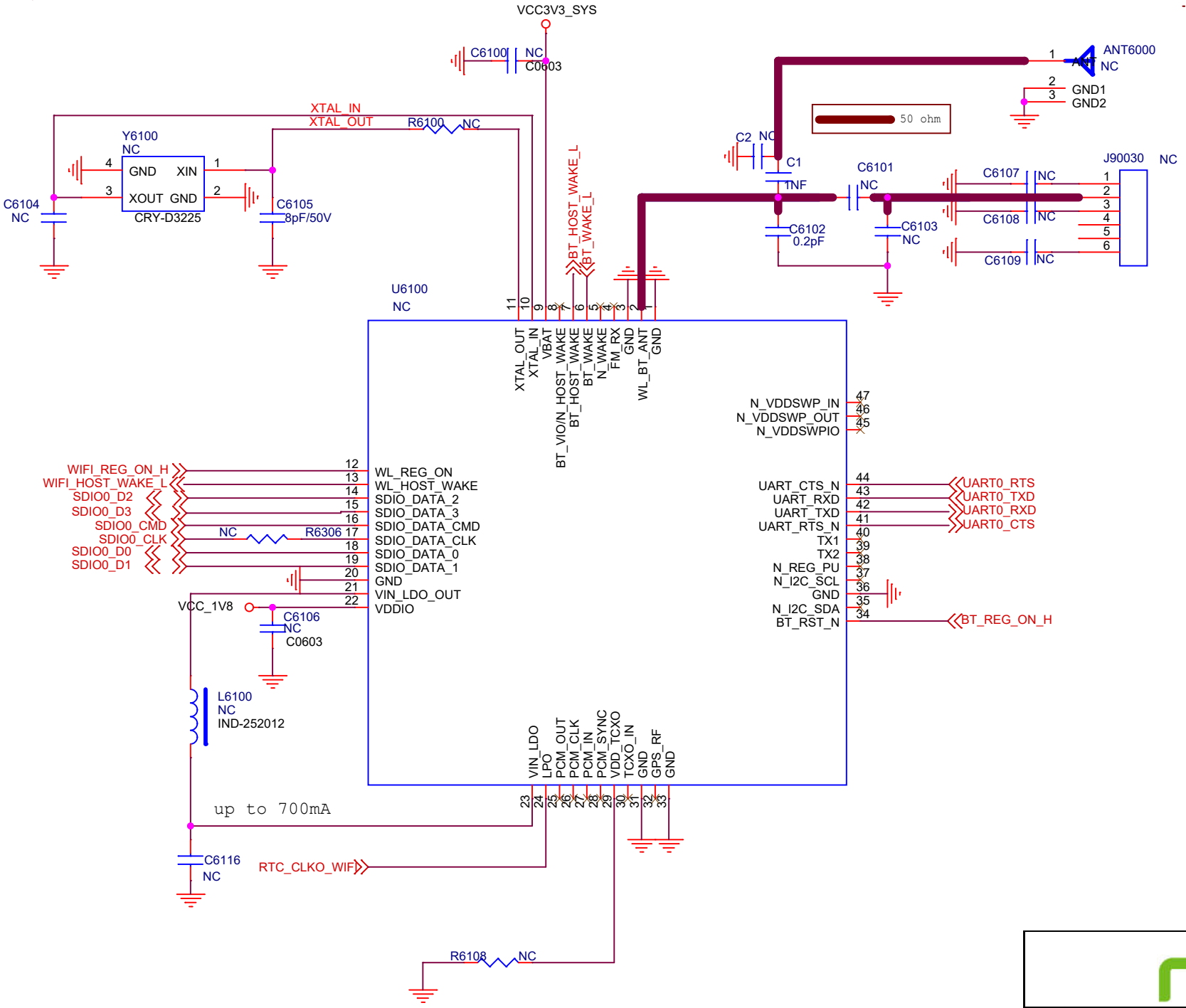
eMMC





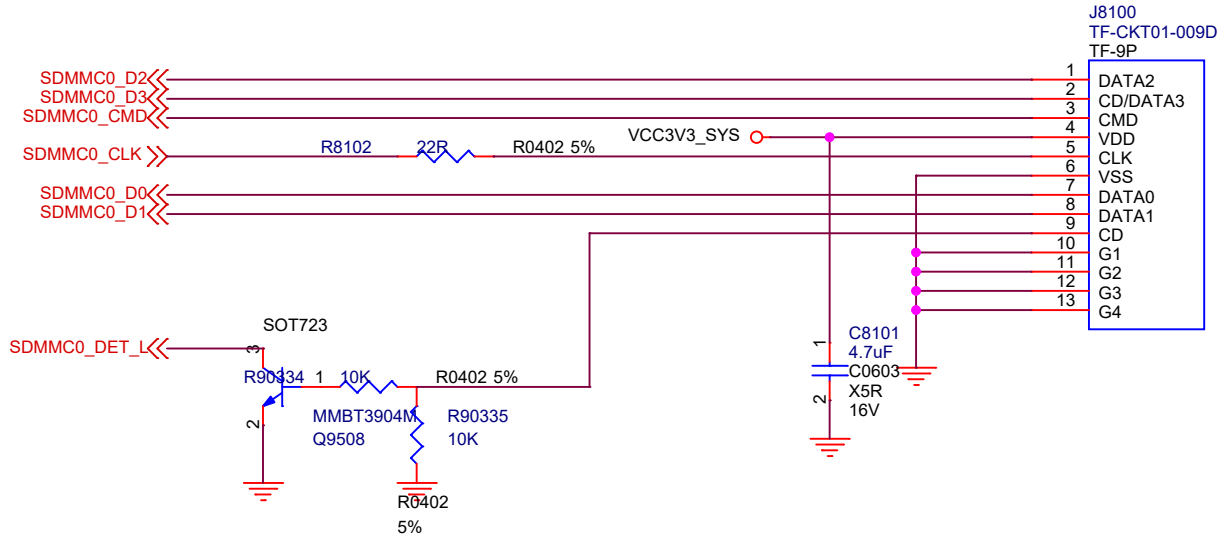


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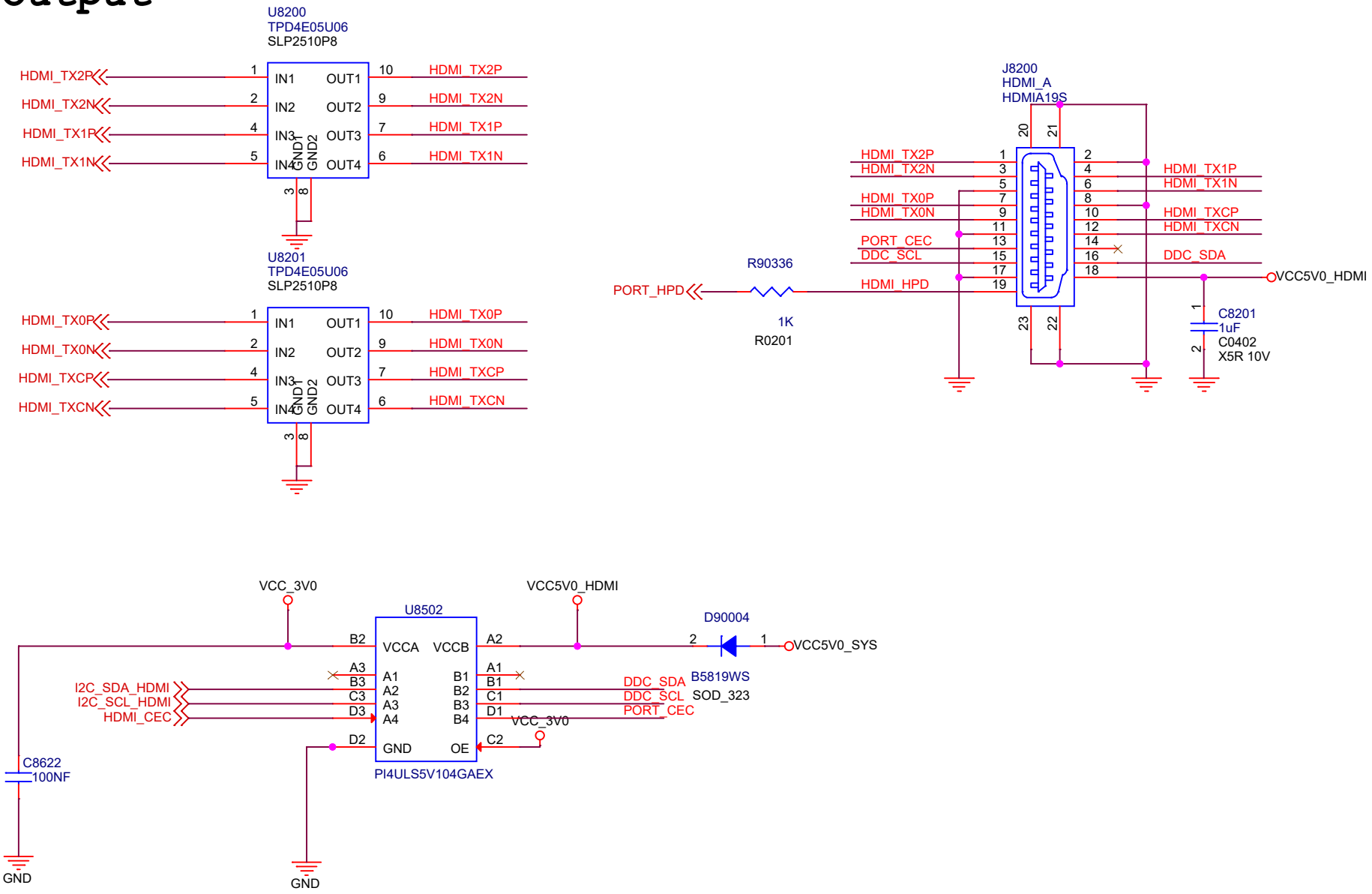
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TF CARD



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HDMI Output



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